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Article

Complementary Bacterial Functions Enhance Mineralization of Aromatic Aliphatic Copolyesters within a Marine Microbial Consortium

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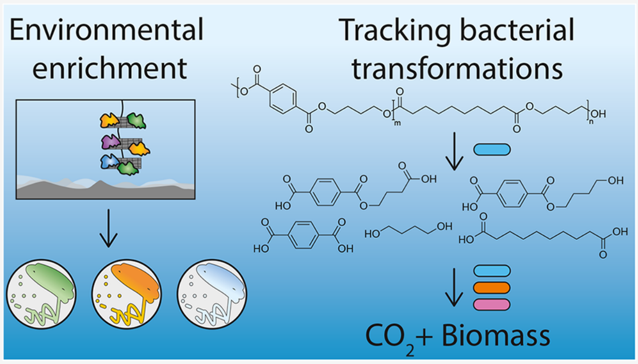
ABSTRACT: Plastics are a major environmental concern due to their persistence in natural systems. Biodegradable plastics can mitigate this impact by reducing their residence time in the environment. To constrain the environmental lifetime of these materials, understanding the fundamental principles dictating their biodegradation is crucial. The work presented here probes this complexity by using a 30-member bacterial community enriched from the marine ecosystem to investigate how bacterial consortia mineralize polybutylene sebacate-*co*-terephthalate (PBSeT), a biodegradable aromatic aliphatic copolyester. Carbon dioxide quantification and isotopic tracing provided evidence of polymer mineralization, while monoculture phenotyping demonstrated no one bacterium could consume all polymer components. Further, coculture incubations revealed complementary functions between community members enhanced mineralization. To explain this enhanced mineralization, dissolved organic carbon and chemical product tracking were performed. Notably, depolymerization of the bulk polymer was dictated by a bacterium unable to consume all polymer components (*Pseudomonas pachastrellae*), requiring complementary bacteria to achieve enhanced mineralization (*Pseudooceanicola nitratireducens* or *Peribacillus frigoritolerans*). This yielded direct experimental evidence of the complementary bacterial transformations that may control polymer mineralization in the environment.



KEYWORDS: *biodegradation, microbial consortia, polyesters, plastics, pollutant fate*

INTRODUCTION

Plastic waste accumulates in natural systems[1] due to slow rates of degradation[2] relative to the rate of input, causing undesirable environmental impacts.[3] For applications where environmental release is inevitable, such as agricultural mulch films, biodegradable polymers provide shorter-lived alternatives[4−6] as microorganisms can mineralize these materials to CO_2.[7] These polymers often contain ester bonds, which are susceptible to hydrolytic enzymes such as lipases and cutinases.[8] Bacteria naturally produce hydrolytic enzymes to degrade ester-containing compounds like cutin,[9] and polyesters are thought to be susceptible to similar degradation processes. Aromatic aliphatic copolyesters are a type of biodegradable polyester that exploit this functional group and are currently used in different applications such as agricultural mulch films,[10−13] biowaste bags, and selected packaging applications. These materials combine aromatic phthalates with aliphatic groups linked by ester bonds. The aromatic portion confers strength and durability comparable to poly(ethylene terephthalate) (PET),[14,15] while the aliphatic groups improve enzymatic accessibility.[16−18] Importantly, this mixed chemistry may introduce greater complexity than commodity plastics. For example, poly(butylene sebacate-*co*-terephthalate) (PBSeT) is an aromatic aliphatic copolyester that contains moieties of terephthalic acid (TPA), sebacic acid (SA), and 1,4-butanediol (Bd). TPA is an aromatic, dicarboxylic acid requiring ring-dioxygenase enzymes to convert TPA into protocatechuic acid, which is subsequently funneled into central carbon metabolism via aromatic ring-cleavage pathways.[19] SA is an aliphatic, dicarboxylic acid that is metabolized via β-oxidation-like pathways.[20] Finally, Bd is an aliphatic diol

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that requires oxidation of the terminal alcohol groups to carboxylic acids before further processing into central carbon metabolites.[21] It is unlikely that a single bacterium can both depolymerize the polymer and metabolize all subunits of an aromatic−aliphatic copolymer. Given this complexity, degradation may require consortia with complementary metabolic functions. Indeed, studies suggest that consortia play a central role in the biodegradation of polyesters.[22−24] However, principles describing the transformation and consumption of these materials by bacteria remain elusive. Most evidence comes from "-omics"-based approaches, where microbial composition cannot be controlled. Previous studies either indirectly investigated the degrader and opportunist species,[22,25,26] tracked chemical products through inference or bulk measurements,[27,28] or have focused on single organisms.[29,30]

Although the degradation of complex biomass in nature is rarely the work of a single species,[31] one well-studied exception is *Ideonella sakaiensis*, a PET-degrading soil bacterium.[29] This organism can achieve complete PET mineralization through a distributed degradation process by producing PETase and MHETase, two complementary enzymes that together depolymerize PET into compounds that the organism can consume.[32,33] PETase acts only on the insoluble polymer, while MHETase acts only on the soluble monomer, and both are required for full degradation.[34−36] By analogy, a similar division of labor may occur across different bacteria within microbial consortia degrading chemically complex biodegradable polymers.[24,31,37]

This type of cooperative degradation has been previously investigated using metagenomics, transcriptomics, and proteomics in mixed microbial communities to classify taxa, identify expressed genes, and detect proteins, respectively.[22,23,26,38] These approaches revealed that distinct bacterial species colonized plastic particles compared to the bulk water column and nonplastic controls.[27,38−42] For example, Meyer-Cifuentes et al. hypothesized that cooperative metabolism contributed to net degradation of aromatic aliphatic copolyesters, showing that these plastics enriched diverse microbial communities.[22] While some studies have directly investigated cultured bacteria living in biofilms attached to plastic particles,[43] no study has provided direct evidence of bacterial interactions leading to enhanced mineralization of bulk polyesters. Indeed, studying bacterial-mediated depolymerization involved in polymer mineralization often requires direct measurement of intermediate degradation products, such as oligomers. Yet, in mixed microbial communities, these intermediates may be rapidly consumed, obscuring detection. This transient and dynamic nature of polymer-associated chemicals limits assignment of a specific step in the degradation process to a particular species or enzyme. Therefore, investigation into cooperative degradation requires work in model systems where the behavior of individual species in monoculture can be contrasted to the behavior of the community.

In this study, we enriched a bacterial community collected from the Mediterranean Sea grown on the commercial blended form of the PBSeT polymer and isolated individual species to test the hypothesis that mineralization of complex polyesters is dependent on distributed degradation. The enriched bacteria spanned Alphaproteobacteria (14), Gammaproteobacteria (13), Bacilli (two), and Flavobacteria (one), and together formed a stable consortium after serial passaging on PBSeT (Figure S1). Mineralization of PBSeT to CO_2 by this community was assayed, including stable isotope tracing. Growth of each strain on polymer substructures and esterase activity was performed to inform selection of coculture incubations for further investigations into how community composition affected mineralization. To contextualize these coculture mineralization assays, dissolved organic carbon (DOC) accumulation was monitored and specific chemical products using liquid chromatography coupled to mass spectrometry (LC−MS) were tracked. Ultimately, this approach sheds light on the complementary bacterial functions leading to transformation and subsequent mineralization of biodegradable polyesters in the marine environment which could help to inform sustainable material design and the development of principles for engineered and natural biological remediation.

■ MATERIALS AND METHODS

Chemicals

PBSeT polymer was sourced from BASF and used as delivered. Analytical grade standards for TPA and SA, and HPLC-grade methanol, as well as the polyester poly(3-hydroxybutyrate) (P3HB) were purchased from Sigma-Aldrich. The esterase activity assay utilized *p*-nitrophenyl butyrate purchased from Sigma-Aldrich. Milli-Q water was generated in-house.

Isolation of Marine Bacteria

Blended plastic films of the polymers polybutylene sebacate-*co*-terephthalate (PBSeT), polybutylene adipate-*co*-terephthalate (PBAT), and polylactic acid (PLA) were incubated with marine sediment and seawater, or with marine sediment only as inoculum. The inoculum from environmental incubations (see Supporting Information for details) was collected by HYDRA Marine Science at the coastal regions of the islands Pianosa (sediment in 40 m depth: 42°34′41.4″N 010°06′30.6″E) and Elba (beach sediment: 42°44′00.1″N 010°09′15.3″E; seawater: 42°44′06.5″N 010°10′33.5″E), Italy, in the Mediterranean Sea. Briefly, more than 700 colonies were picked from enrichment plates inoculated from the biodegradation test samples. These isolated organisms were mixed and serially passaged on the PBSeT polymer for 5 weeks. At 5 weeks, metagenomics was performed on the samples, and the most abundant organisms were identified (Figure S1; $n = 30$). These organisms made up the synthetic community used in this report.

16S Identification of Bacterial Isolates and Whole Genome Sequences

The 30 isolates were grown in rich media (Difco 2216 Marine Broth) overnight. An aliquot (2 μL) of each culture was used for colony polymerase chain reaction (PCR). The 16S rRNA gene was amplified using the reverse and forward primers 1492R (5′-TACGGYTACCTTGTTACGACTT-3′) and 27F (5′-AGAGTTTGATCMTGGCTCAG-3′) (Integrated DNA Technologies), respectively. The PCR products were analyzed via agarose gel electrophoresis to confirm the expected base pair length for the 16S rRNA gene. The products were sent to Azenta and Sanger sequenced using an ABI 3730xl DNA Analyzer. The sequences were then put through a BLAST search against the NCBI rRNA sequence database to identify the organisms.[44]

Whole genome sequences of each organism within the five-member community were obtained by sending purified DNA to SeqCenter (Pittsburgh, PA). Purified genomic DNA was obtained by extracting overnight cultures following the DNeasy Blood and Tissue Kit (Qiagen). SeqCenter performed sequencing, assembly, and gene annotation using an Illumina NovaSeq X Plus sequencer. All genomes and 16S sequences were deposited in GenBank under the accession numbers PRJNA1274597 and PV520552-PV520581, respectively. Whole genomes were screened for the PETase enzyme[44] using tblastn. The sequence for PETase came from the UniProt entry for PETH_PISS1 from *Ideonella sakiensis* under the accession number

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A0A0K8P6T7.[29] Protein structure alignment and statistics were calculated using TM-align, with the crystal structure of PET hydrolase from *I. sakaiensis* (PDB ID: 6ANE)[45] and the AlphaFold predicted structure of the PETase-like enzyme from *Pseudomonas*.

16S Amplicon Sequencing

The 30 bacteria were inoculated into minimal media containing 1.33 mg/mL of PBSeT, P3HB, or no plastic at equal cell concentrations as determined by OD measurements. Five days after inoculation, samples (15 mL) were filtered through cellulose acetate 0.2 μM-filters and prepared for 16S amplicons sequencing. Filters were processed and cells lysed using the DNeasy PowerSoil Pro kit. DNA was extracted from these samples using the DNeasy blood and tissue kit. The DNA concentration (representing the whole sample: particle-attached and free-living bacteria) was quantified using a Thermo Scientific Nanodrop spectrophotometer. DNA samples were sent to SeqCenter (Pittsburgh, PA, USA) for sequencing. Samples were prepared by amplifying the V3/V4 region of the 16S gene using the primers 341F (CCTACGGGDGGCWGCAG, CCTAYGGG-GYGCWGCAG) and 806R (GACTACNVGGGTMTCTAATCC). Analysis was performed on a P1 600cyc NextSeq2000 Flowcell. The data was cleaned and processed using the DADA2 pipeline.[46] Cutadapt was used to remove primers and to trim the reference Sanger sequences of the 30 individual isolates to the V3/V4 region. All samples had more than 100,000 reads analyzed. PBSeT and P3HB were done in duplicate, and the no-plastic control was done in triplicate. The inoculum condition resulted in one useable sample with over 100,000 reads out of the triplicates sent (most likely because of low DNA concentrations). Bacterial identification was classified by comparing the 16S amplicon sequences to the reference 16S sequences for the 30 bacterial isolates used in the experiment.

Laboratory Incubations with Plastic Substrates and the Synthetic Community

For each experiment with the enriched community, individual isolates were taken from cryopreservation in dimethyl sulfoxide (DMSO) at −80 °C, struck onto a rich media plate (Difco 2216 Marine Broth) mixed with agar (Difco), and grown overnight. Individual colonies were picked from each plate and then grown in liquid rich media (Difco 2216 Marine Broth) overnight. The organisms were transferred at least once in the rich media before any experiment. To the best of our ability, each isolate was extracted during the exponential growth phase (as measured by optical density) for use in experimentation. For all incubations, a defined minimal media with slight adaptations was used (see Supporting Information). HEPES buffer was replaced with a bicarbonate buffer system (~6 mM) to more closely mimic the buffer system in the ocean and to remove any interference from potential mineralization of the HEPES buffer. The pH was adjusted to pH 8 using NaOH and/or HCl. For mineralization experiments, all bacterial isolates were mixed in equal starting cell concentrations such that the starting optical density (OD) of the whole inoculum in the sample was 0.01 (approximately 10^7 cells/mL). For DOC accumulation experiments, each bacterium was inoculated at 0.01 OD to control the amount of mobilized DOC from *Pseudomonas*'s depolymerization of PBSeT. For all incubations, the bottles were incubated at lab room temperature (approximately 18 ± 2 °C) and rotated to ensure that the plastic was in contact with the media for the duration of the experiment. To avoid abiotic photodegradation, all bottles were wrapped in aluminum foil.

Dissolved Organic Carbon Analysis

Dissolved organic carbon (DOC) was analyzed on an Elementar Vario Cube total organic carbon analyzer. All samples were acidified with one drop of concentrated HCl and purged with CO_2-free air to remove dissolved inorganic carbon. Potassium hydrogen phthalate standards were prepared before each run and used as a calibration.

CO_2 Measurements

To quantify mineralization of the polymer, evolved CO_2 and its isotopic composition (δ^{13}C) was measured via cavity ring down spectroscopy (CRDS) using a Picarro G-2131i coupled to an AutoMate FX prep device in incubations where the plastic was the sole source of carbon. All incubations were carried out in gastight, 12 mL exetainer vials (Labco) with 4 mL of defined minimal media (see Supporting Information for complete recipe) and approximately 5 mg of plastic (PBSeT) or approximately 2 mg plastic (P3HB). For each measurement, to terminate the bacterial growth as well as acidify the sample to drive CO_2 into the headspace, 2 mL of 10% phosphoric acid was added via the cap septum using a syringe fitted with a hypodermic needle. Samples were run within 24 h of acidification. CO_2-free air was bubbled into the liquid media to act as a carrier into the CRDS, via a Liaison universal interface.[47] The measurements of CO_2 and δ^{13}C were calibrated using natural dissolved inorganic carbon (DIC) taken from the environmental systems lab at woods hole oceanographic institution (WHOI), corrected to the carbonate standard, Harding Iceland Spar (standardized value of 1.5‰), several known isotopic standards (NBS-19, TIRI-F, and IAEA-C2), and the certified reference material provided by Andrew Dickson's laboratory (at Scripps). With each set of 30 samples, a standard set was positioned at the beginning, middle, and end. This rigorous standard protocol allowed for comparison across different analysis days without concern for experimental drift, which was particularly important for long-term incubations (42 days) where the incubation was started at day 0 and time points were sacrificed once per week.

Percent mineralization was calculated by taking the difference between CO_2 measured in the samples containing plastic and plastic-free controls and dividing by the theoretical maximum amount of CO_2 that could have been produced (eq 1). The theoretical maximum CO_2 was defined as the mass of the polymer added to the test vials, multiplied by the carbon content of the polymer (measured on a total organic carbon analyzer).

$$\%\ \text{mineralization} = \frac{CO_2\ \text{samples (mg)} - CO_2\ \text{control (mg)}}{\text{theoretical maximum amount of } CO_2\ \text{(mg)}} \times 100 \quad (1)$$

Isotopic Mass Balance

To determine the isotopic value of respired carbon in degradation experiments where P3HB or PBSeT were present, the use of isotopic mass balance was employed following this equation

$$\delta^{13}C_{\text{sample}} = \delta^{13}C_{\text{control}} \times f_{\text{control}} + \delta^{13}C_{\text{polymer}} \times f_{\text{polymer}} \quad (2)$$

$$f_{\text{control}} = \frac{CO_2\text{control(mg)}}{CO_2\text{sample(mg)}} \quad (3)$$

$$f_{\text{polymer}} = \frac{CO_2\text{sample(mg)} - CO_2\text{control(mg)}}{CO_2\text{sample(mg)}} \quad (4)$$

where f_{control} is the fraction of measured CO_2 coming from inoculum carbon respiration in control bottles with no plastic added (this encompassed respiration of any marine broth from the inoculum and the bicarbonate used as a buffer system), f_{polymer} is the fraction of measured CO_2 coming from polymer respiration, $\delta^{13}C_{\text{control}}$ is the isotopic signature of control bottles (i.e., the measured value of control samples with no plastic added), $\delta^{13}C_{\text{polymer}}$ is the calculated isotopic signature of the CO_2 from polymer respiration (calculated via eq 2), and $\delta^{13}C_{\text{sample}}$ is the isotopic signature of the entire measured CO_2 (raw isotopic data are available in the Supporting Information, Figure S2). This equation was then solved for the $\delta^{13}C_{\text{polymer}}$ to determine the isotopic signature of CO_2 from polymer respiration. The isotopic signatures of the bulk starting materials P3HB, PBSeT, rich media (Difco 2216 Marine Borth), and bicarbonate were measured by isotope ratio mass spectrometry (IRMS) at the UC Davis Stable Isotope Facility.

Plate Reader Growth Experiments

Growth experiments of each isolate with the chemical components of the polymer were conducted using a BioTek Epoch2 microplate reader. Minimal media was made as described above with the addition of 15 mM TPA, SA, or Bd. TPA and SA were first dissolved in water

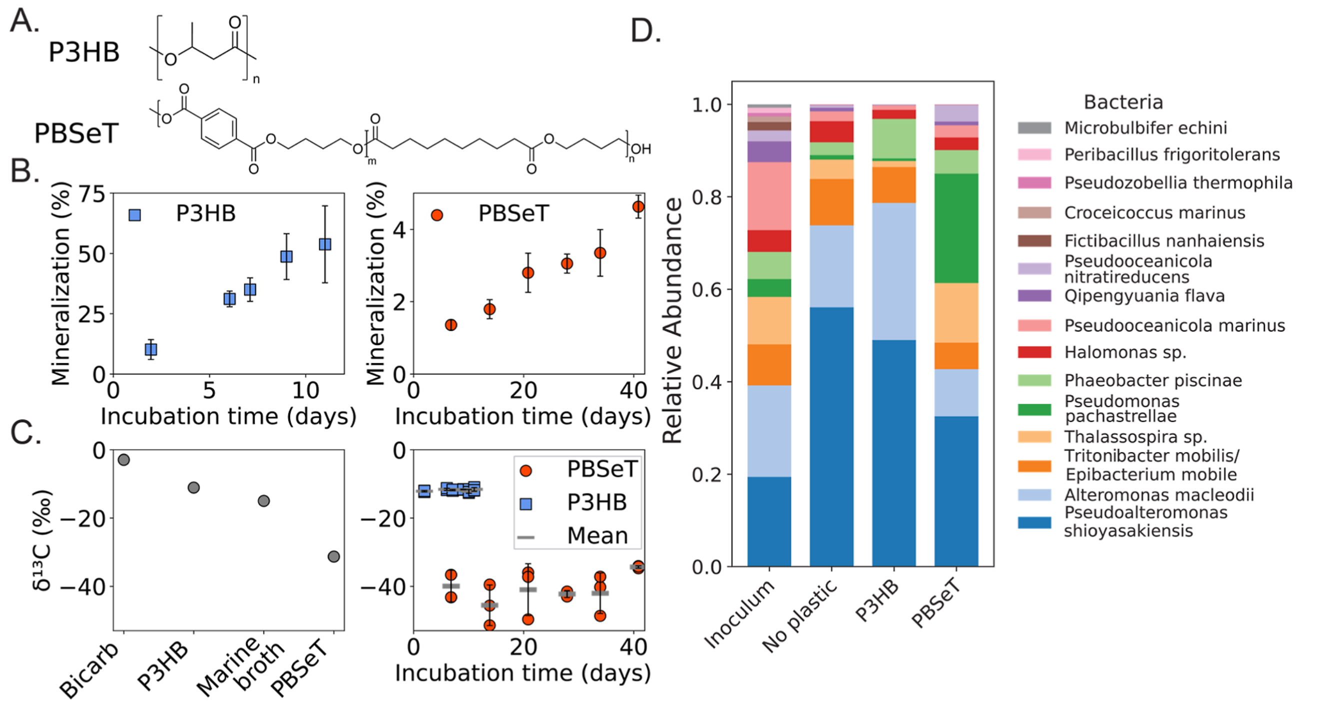

Figure 1. Mineralization of the polymers P3HB and PBSeT to CO_2 by the Mediterranean Sea-derived community, natural-abundance isotopic tracing, and bacterial community composition. (A) The chemical structures of the polymers P3HB and PBSeT. (B) Mineralization measured as respired CO_2 from the 30-strain bacterial community grown with P3HB (left) and PBSeT (right) as the sole source of carbon. P3HB was respired within 11 days, whereas PBSeT was partially respired in 42 days. Error bars represent standard deviation across biological triplicate incubations. (C) Natural abundance isotopic signatures of the bulk materials (P3HB and PBSeT), bicarbonate buffer, and inoculum carbon (marine broth) in bacterial incubations (left). Error bars on triplicate isotopic analyses are smaller than the symbol. Natural abundance isotopic signatures of the respired CO_2 in polymer-only incubations (right). Error bars represent standard deviation from three biological replicates, and horizontal bars illustrate the mean of the three replicate inoculations. (D) 16S amplicon sequencing at day five of an incubation with the 30 bacterial species and the polymers P3HB, PBSeT, or no plastic control. The inoculum represents the composition of the microbial community at time 0.

(at approximately 100 mM) and raised to pH 10 to completely dissolve the compounds, diluted into the media to reach a final concentration of 15 mM, and phosphoric acid (10% v/v) was added dropwise to pH 8. After this adjustment, the solution remained clear and was used for incubations. For the experiments, bacteria were inoculated into a 96-well plate to a starting OD of 0.01, incubated with shaking at room temperature, and growth was tracked as optical density with absorbance at 500 nm (OD500). When reporting the OD measurements, the absorbance values of the no additional carbon controls were subtracted from each experimental sample to minimize apparent contributions from growth on background carbon.

Esterase Assay

To quantify the activity of extracellular esterases, a protocol was adapted from Inglis et al. for a colorimetric assay that uses the hydrolysis of the molecule *p*-nitrophenyl butyrate (PNB) to the colored compound *p*-nitrophenol (λ_{max} 405 nm)[9] as a proxy for enzyme activity. Extracellular esterases, cutinases, or hydrolases can catalyze this hydrolysis and so enhance the degradation of PNB into *p*-nitrophenol. Bacterial culture grown in rich media (Difco 2216 Marine Broth) (1 mL) was centrifuged at 10,000 G for 2 min to remove microbial cells and biomass. The supernatant (20 μL) was transferred to a 96-well plate containing 180 μL of 2.7 mM PNB in a phosphate buffer system at pH 7.5. The 96-well plate was incubated for 30 min at 25 °C with shaking, and absorbance was read at 405 nm.

Targeted Chemical Analysis of Degradation Products

An Agilent Infinity II high performance liquid chromatography (HPLC) system coupled to an agilent triple quadrupole mass spectrometer (QqQ-MS) with an electrospray ionization (ESI) source in negative mode was used to detect the production of TPA and SA containing degradation products. The mobile phases used were LC−MS grade methanol (B) and LC−MS grade water (A) with 0.1% formic acid purchased from Sigma-Aldrich. The method employed was: 98% A 2% B for 2 min and diverted to waste so as not to introduce salts to the MS, the solvents were ramped to 70% A:30% B from 2 to 2.1 min, 20% A:80% B from 2.1 to 3.35 min, to 2% A:98% B form 3.35−4.5 min, and held until 5.5 min. The mass transitions monitored were TPA (precursor m/z = 165, product m/z = 121), SA (precursor m/z = 201, product m/z = 189), 4-((4-hydroxybutoxy)carbonyl) benzoic acid (BdTPA, precursor m/z = 237, product m/z = 121), and 4-((3-carboxypropoxy)carbonyl) benzoic acid (BdTPAOx, precursor m/z = 251, product m/z = 165). All peak areas were normalized to the internal standard ^{13}C-labeled TPA (MedChem Express). SA and TPA were further quantified with authentic standards. Representative extracted ion chromatograms can be found in the Supporting Information (Figure S3).

RESULTS AND DISCUSSION

Mineralization of PBSeT and P3HB

The 30 bacterial species were grown with either PBSeT or the polyester P3HB, and CO_2 production was quantified to assess their ability to mineralize these polymers (Figure 1A,B). P3HB, which is known to biodegrade in marine environments,[48−50] was used as a reference polymer to contextualize the potential functional specificity of PBSeT mineralization by these bacteria. The polymers were the sole source of carbon besides the small amount of carryover carbon from the inoculum. Polymer mineralization was accounted for by control bottles where no plastic was added and confirmed through natural abundance isotopic tracing. The enriched community mineralized 53.8 ± 16.0% of P3HB after 11 days and 4.6 ± 0.3% of PBSeT after 41 days. Polyethylene controls showed 0%

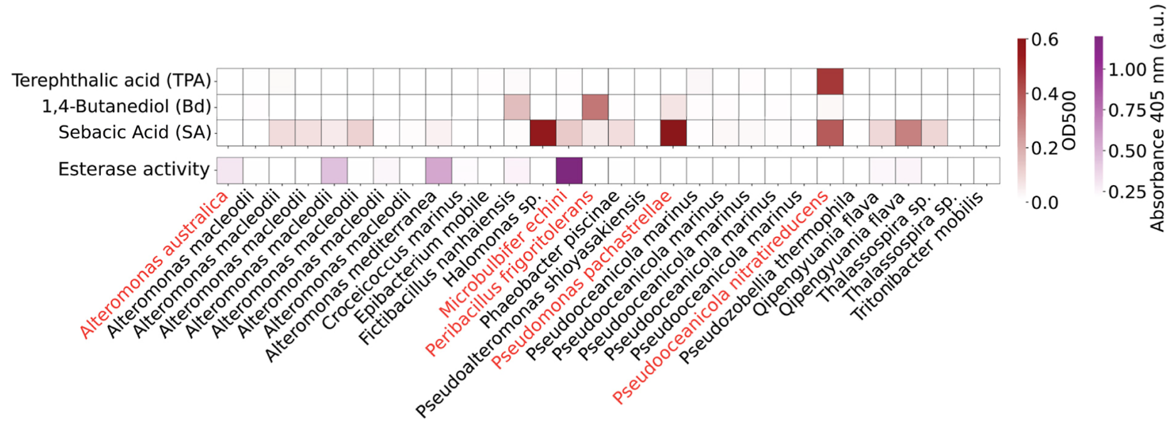

Figure 2. Individual bacteria exhibited diverse abilities to mobilize and consume PBSeT chemical components. Growth of each bacterium on polymer components (TPA, Bd, or SA), measured via optical density (OD) at the end of a 24 h incubation (red heat map). Esterase activity (purple heatmap) was assayed using a colorimetric probe molecule, *p*-nitrophenyl butyrate. All data shown are averages of at least two biological replicates. Species names were determined by 16S profiling. Bacteria highlighted in red indicate organisms used for further experimentation and were chosen due to their ability to either consume TPA, SA, or Bd, or their ability to produce esterases.

mineralization (Figure S4). Isotopic tracing further confirmed the mineralized CO_2 was associated with polymer respiration and not respiration of background carbon. The natural abundance ^{13}C composition of evolved CO_2 was compared to that of the bulk materials' isotopic signatures (Figure 1B). Briefly, carbon ^{13}C isotopic signatures of a metabolite should reflect the isotopic signature of the source substrate with minor fractionation effects (±1‰),[51] allowing for carbon tracing during metabolism. Using a mass balance approach to disentangle the isotopic composition of polymer derived CO_2, P3HB-derived CO_2 had an average isotopic composition of −11.79 ± 0.40‰ across the incubation time. The bulk P3HB was −11.06 ± 0.09‰, a statistically significant difference (p-value <0.00005) from the $\delta^{13}C$ of P3HB-derived CO_2, but within anticipated metabolic fractionation effects.[52] Thus, the $\delta^{13}C$ of CO_2 reflected metabolism of the P3HB bulk material by the bacterial community. Similarly, the $\delta^{13}C$ of CO_2 from incubations with PBSeT showed signatures consistent with PBSeT mineralization. The respired CO_2 from these incubations exhibited a stable carbon isotopic signature of −41 ± 6‰, averaged across all incubation times (Figure 1B), in comparison to −31.3 ± 0.2‰ for the PBSeT starting material. This depletion could be due to preferential utilization of some pieces of the polymer relative to others as PBSeT is made of three distinct chemical subunits (see Supporting Information for more discussion on PBSeT derived CO_2 isotopic signatures). Further, the stable carbon composition of the rich media (Difco Marine Broth 2216, present only as a small fraction due to small carry over from the inoculum, −14.96 ± 0.05‰) and bicarbonate buffer (−2.91 ± 0.04‰) were enriched relative to the respired CO_2 and so could not explain the depleted signature of the evolved CO_2 in PBSeT incubations. Thus, the mineralization measured must have reflected metabolism of the plastic rather than an artifact of background carbon consumption. This supports that the community was consuming PBSeT polymer as a carbon and energy source.

To identify which bacterial species were enriched in response to PBSeT and P3HB degradation, 16S amplicon sequencing was used to trace the 30-member bacterial community over a 5 day incubation with each polymer. Importantly, DNA was extracted from the entire filtered sample to retain both planktonic and particle-attached bacteria. The relative abundance of two taxa increased in P3HB conditions relative to the no plastic controls: *Alteromonas macleodii* (17.7 ± 0.6% to 29.7 ± 0.9%, p-value = 0.0004) and *Phaeobacter piscinae* (2.8 ± 0.2% to 8.6 ± 0.1%, p-value = 0.00003). For PBSeT conditions, the relative abundance of four taxa increased in PBSeT conditions relative to the no plastic controls: *Pseudomonas pachastrellae* (0.96 ± 0.26% to 23.7 ± 0.5%, p-value = 0.000002), *Pseudooceanicola nitratireducens* (0.4 ± 0.06% to 3.5 ± 0.9%, p-value = 0.004), *Thalassospira*sp. (4.2 ± 0.2% to 12.9 ± 0.9%, p-value = 0.0002), and *P. piscinae* (2.8 ± 0.2% to 5.1 ± 0.05%, p-value = 0.0003) (Figure 1D). DNA concentrations across conditions supported the sequencing results, showing higher biomass in polymer-containing incubations. DNA concentrations of a 100 μL eluent were 288 ± 13, 69 ± 5, 40 ± 7, and 4.13 ± 0.06 ng/μL measured on a nanodrop for incubations with P3HB, PBSeT, no carbon, and the inoculum, respectively. The higher DNA yields confirmed bacterial growth on P3HB and PBSeT, and sequencing highlighted *P. pachastrellae*, *Pseudooceanicola nitratireducens*, *Thalassospira* sp. and *P. piscinae* as plausible contributors to PBSeT degradation.

Growth Assays on the Subunits of PBSeT

Monoculture growth assays on PBSeT subunits revealed the chemical components of the polymer utilized by each bacterium. Single isolates were tested for growth on the subunits of PBSeT as a sole carbon source (TPA, SA, or Bd; chosen to mimic the simplest form of polymer-associated carbon available during degradation). Based on these assays, 14 bacteria showed some growth on SA, three of which also grew on Bd (*Fictibacillus nanhaiensis*, *Peribacillus frigoritolerans*, and *Pseudomonas pachastrallae*). Only one organism grew on TPA (*Pseudooceanicola nitratireducens*), consistent with the expectation that phthalate-degrading capabilities require specialized enzymes, not widely distributed among bacteria.[53,54] A subsequent extracellular esterase assay was performed to determine bacteria implicated in the depolymerization of the polymer. Extracellular esterases or other hydrolases (similar to PETase), have been shown to degrade polyesters outside the cell.[8] Thus, the organisms with extracellular esterase activity may play a key role in bulk polymer depolymerization prior to oligomer or monomer transformations and/or consumption.

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Esterase activity was determined using the probe molecule *p*-nitrophenyl butyrate. Activity was observed in nine organisms (*Alteromonas australica*, two *A. macleodii* strains, *Alteromonas mediterranea*, *Croeicoccus marinus*, *Fictibacillus nanhaiensis*, *Microbulbifer echini*, and two *Qipengyuania flava* strains; Figure 2). *M. echini* produced significantly higher extracellular esterase activity than the other isolates (p-value <0.05). Notably, no single organism combined extracellular esterase production and consumption of all polymer subunits, implying consumption of the PBSeT polymer necessitated multiple bacteria.

Guided by these results, a pared-down community was constructed and used to test the effect of community composition on mineralization (Figure 2, red highlighted species). *P. pachastrallae*, *Pseudooceanicola nitratireducens*, and *Peribacillus frigoritolerans* were selected for their strong growth on SA, TPA, and Bd, respectively. Additionally, *P. pachastrallae* and *Pseudooceanicola nitratireducens* were abundant during PBSeT degradation measured via 16S amplicon sequencing (Figure 1D) and so, may have an important role in PBSeT degradation. While *Halomonas*sp. grew well with SA, this organism did not show a significant abundance change and so was excluded from the minimized community. *M. echini* was included for its high esterase activity, and *A. australica* was selected as a negative control bacterium, exhibiting little esterase activity and no ability to grow on TPA, SA, or Bd.

Mineralization of PBSeT was Controlled by *P. pachastrellae*

The strategically selected five-member community replicated the original 30-member community's mineralization capability (Figure 3A). After 42 days, the selected bacteria mineralized 5.6 ± 1.1% of added PBSeT as compared to the 30-member community mineralizing 4.6 ± 0.3% (not statistically different, p-value = 0.34). Interestingly, the selection was specific for the

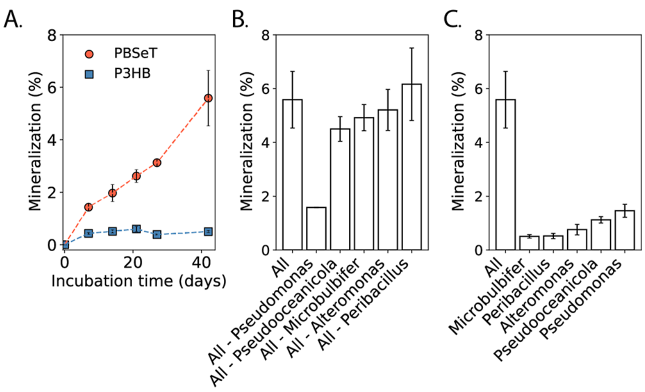

Figure 3. Mineralization of PBSeT to CO_2 was replicated by a five-member bacterial community selected for specific metabolic capabilities. (A) The five-strain bacterial community grown with PBSeT (red) or P3HB (blue) as the sole source of carbon. The five-strain community exhibited the same mineralization capability toward PBSeT as the 30-strain community but lost the ability to consume P3HB. (B) Species "knock-out" incubations with PBSeT as the sole source of carbon, measured at 42 days of incubation. Taking *Pseudomonas* out of the five-member community showed the largest decrease in mineralization; here, the error bar on biological triplicates is too small to visualize. (C) Monoculture incubations with PBSeT as the sole source of carbon, measured at 42 days of incubation. Mineralization of PBSeT decreased for all monoculture incubations as compared to the mixed community. All error bars indicate standard deviation from biological triplicates.

PBSeT polymer, as the five-member community lost the ability to grow on the P3HB polymer. This is not entirely unexpected, since depolymerization of P3HB requires specific PHB-depolymerases[49,55,56] that were not explicitly selected here, implying enzymatic depolymerization of PBSeT is distinct from that of P3HB. Thus, the chosen bacteria were metabolically specialized to degrade the PBSeT polyester. This is consistent with previous "-omics" studies, which illustrated chemically distinct polyesters enrich distinct microbial communities.[23]

To provide a quantitative measure of complementary metabolisms affecting polymer mineralization, we monitored impacts on the evolved CO_2 while eliminating single organisms from the five-member minimal community and quantifying CO_2 at a single 42 day time point (herein referred to as species "knock-out" experiments). A key bacterial member in the community, *P. pachastrallae*, exhibited the largest impact on mineralization (Figure 3B). Omitting *Pseudomonas* from the community decreased mineralization from 5.6 ± 1.1% to 1.58 ± 0.01% (significantly different; p-value = 0.022). In contrast, omitting *Pseudooceanicola*, *Microbulbifer*, *Alteromonas*, and *Peribacillus* did not show statistically significant changes (p-values >0.05). *Pseudomonas*'s control on mineralization was somewhat unexpected given that this organism did not exhibit appreciable esterase activity. In contrast, removal of the highest esterase-producing organism, *Microbulbifer*, did not significantly reduce mineralization. Although extracellular esterases from *Microbulbifer* may contribute to depolymerization of PBSeT, the organism's inability to metabolize polymer-associated degradation products (Figure 2) could explain its limited role in overall mineralization. To explain *Pseudomonas*'s control on mineralization we hypothesized that this organism produced an extracellular enzyme capable of depolymerizing PBSeT that was not detectable by the esterase assay.

To test this hypothesis, the genomes for each of the five organisms were screened for polyester-degrading enzymes. Using the amino acid sequence for the PETase enzyme from *I. sakaiensis* as a query, a tBLASTn search across the five genomes revealed only one positive hit with an e-value of 4.38×10^{-66} and sequence identity of 48% at the amino acid level (see Figure S5 for visual alignment with the Alphafold predicted structure[57]). In agreement with the species knockout experiments, this match was in the genome for *Pseudomonas*, suggesting this organism encoded a polyester-degrading enzyme which it may secrete to depolymerize PBSeT. The lack of detectable esterase activity in *Pseudomonas* (Figure 2) may reflect several possibilities: (1) the conditions of the original esterase assay did not induce expression of *Pseudomonas*'s PETase-like enzyme; (2) this enzyme was membrane-bound and therefore absent from the filtrate used in the assay; or (3) *p*-nitrophenyl butyrate is a poor probe molecule for detecting PBSeT-depolymerizing enzymes. PETase-like-enzymes were also identified and screened for activity in the complex community samples of Meyer-Cifuentes et al.,[22−24,37] showing these enzymes may be influential in copolyester degradation in the environment. While no other organism of the five tested contained an enzyme similar to PETase, *Pseudooceanicola* uniquely carried the genes pcaG and pcaH. These genes encode subunits of the enzyme 2,3-protochuate dioxygenase, which is known to be involved in the degradation of phthalates.[54,58,59] This supports the observation that *Pseudooceanicola* was uniquely able to consume TPA,

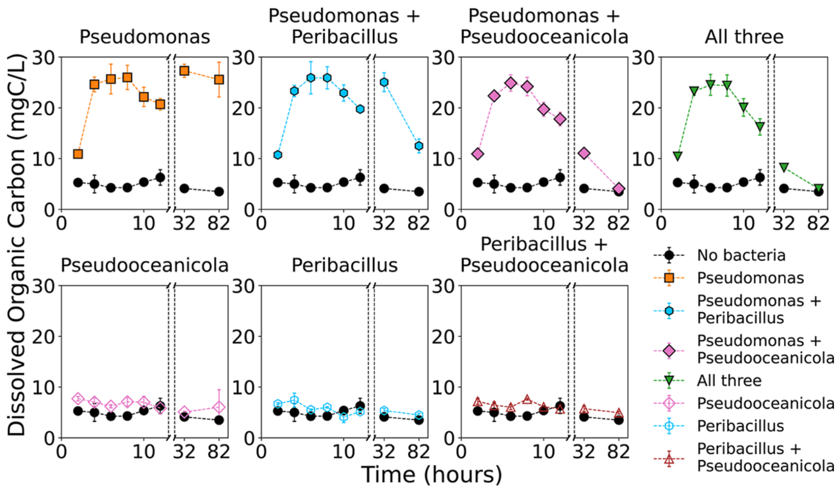

Figure 4. Accumulated PBSeT-derived DOC from *Pseudomonas* was available to *Pseudooceanicola* and *Peribacillus*. Cocultures of *Pseudomonas*, *Pseudooceanicola*, and *Peribacillus* were incubated with 6.25 mg/mL of PBSeT and DOC measured using a total organic carbon analyzer. DOC accumulated in all *Pseudomonas* conditions and did not accumulate in conditions lacking *Pseudomonas* (i.e., *Peribacillus*, *Pseudooceanicola*, and *Peribacillus* + *Pseudooceanicola* conditions). DOC concentrations showed a transient accumulation and decreased with time in the presence of *Peribacillus* and *Pseudooceanicola* (i.e., the *Pseudomonas* + *Peribacillus* and *Pseudomonas* + *Pseudooceanicola* conditions).

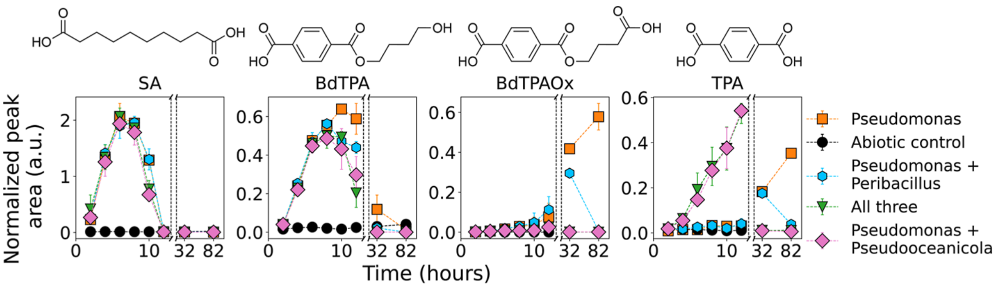

Figure 5. Chemical transformations of polymer-derived DOC by *Pseudomonas*, *Pseudooceanicola*, and *Peribacillus*. A targeted LC−MS analysis monitoring the compounds SA (m/z = 201), BdTPA (m/z = 237), BdTPAOx (m/z = 251), and TPA (m/z = 165) was performed to assess accumulation and removal of chemically specific compounds by each bacterium. All peaks were normalized to the internal standard ^{13}C TPA. In all incubations with *Pseudomonas*, SA was removed (consistent with the organism's ability to consume this compound), while phthalate-containing compounds accumulated. In coculture with *Pseudooceanicola* and *Peribacillus*, phthalate containing compounds were removed. Additionally, *Pseudooceanicola* was able to hydrolyze phthalate-containing oligomers into TPA and subsequently consume this TPA.

highlighting its potentially significant role in mineralization of PBSeT-derived DOC.

Indeed, each bacterium on its own was unable to efficiently mineralize the polymer, indicating complementary bacterial functions between a depolymerizer and other species consuming the mobilized products were required to promote polymer carbon mineralization. Specifically, when grown in monoculture, each bacterium showed a significant decrease in mineralization as compared to the whole community (Figure 3C, p-values <0.05). *Pseudomonas* showed the highest mineralization (1.5 ± 0.2%) but did not reach the mineralization of the entire five-member community (5.6 ± 1.1%). Previous work has shown that accumulation of degradation products can inhibit plastic-degrading enzymes, such as PETase.[30,31] Since *Pseudomonas* cannot consume TPA-containing products, inhibition of the PETase-like-enzyme through accumulation of these dissolved compounds may have occurred in the monoculture experiment. This highlights the potential complementary metabolic roles each bacterium played in mineralization of PBSeT.

PBSeT DOC Accumulation and Consumption

Importantly, the species "knock-out" and monoculture experiments uncovered complementary bacterial functions otherwise difficult to observe in polymer biodegradation studies. Typical "-omics" studies rely on mixed microbial communities to understand how polymer carbon is shared among bacteria, with limited control over community composition and ability to track transient products.[22,42,60,61] Consequently, claims that microbial interactions enhance mineralization are often inferential rather than mechanistic. The model system developed in this study revealed that while mineralization was controlled by *Pseudomonas*, enhanced mineralization necessitated the involvement of additional bacterial species.

To further investigate this carbon sharing among these bacteria, DOC accumulation and removal was analyzed by total organic carbon analysis and liquid chromatography coupled to mass spectrometry (LC−MS) in cocultures of *Pseudomonas*, *Peribacillus*, and *Pseudooceanicola* grown with PBSeT (Figure 4). These three bacteria were chosen for their distinct capabilities; *Pseudomonas* was hypothesized to depolymerize the polymer and consume SA; whereas *Pseudooceanicola* was hypothesized to consume TPA; and *Peribacillus* was hypothesized to consume Bd. In all incubations containing *Pseudomonas*, DOC accumulated, reaching a maximum of ∼25 mgC/L between 6−8 h. In contrast, incubations lacking *Pseudomonas* showed no DOC accumu-

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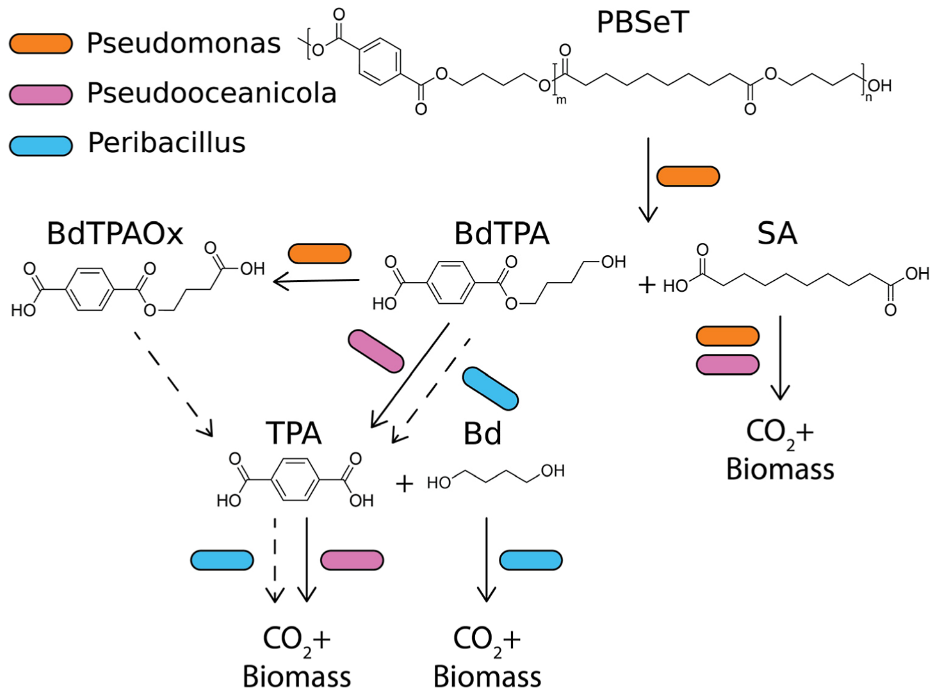

Figure 6. Metabolism of PBSeT relied on distributed degradation by the enriched marine microbial community. Proposed route of PBSeT degradation by the characterized bacteria. Solid arrows indicate pathways with high confidence via direct experimental evidence, whereas dashed arrows are hypothesized or supported via indirect evidence. *Pseudomonas* controlled depolymerization, predominantly mobilizing the polymer into SA and BdTPA. It further oxidized BdTPA into BdTPAOx and consumed SA as its carbon source. *Pseudooceanicola* hydrolyzed phthalate containing oligomers into TPA and consumed this product. Additionally, *Pseudooceanicola* could consume SA as determined by growth assays. Finally, *Peribacillus* could consume Bd and showed the ability to remove phthalate-containing compounds.

lation, supporting the hypothesis that *Pseudomonas* controlled mineralization via PBSeT depolymerization (see Supporting Information for controls of the bacteria with no plastic Figure S6). Further, in the *Pseudomonas*-only condition, DOC levels decreased from their maximum at 6−8 h through to 12 h, plausibly due to *Pseudomonas* consumption. An increase in DOC was observed at 32 h which persisted through the end of the incubation. This accumulation of DOC from 32 to 82 h could arise from *Pseudomonas*'s inability to consume all subunits of PBSeT (Figure 2).

With addition of *Peribacillus* and *Pseudooceanicola*, DOC initially accumulated between 2 and 8 h similar to the *Pseudomonas*-only condition but showed removal over longer time scales. For both cocultures, from 32 to 82 h DOC decreased relative to the *Pseudomonas*-only condition, showing that both *Peribacillus* and *Pseudooceanicola* contributed to the removal of polymer DOC mobilized by *Pseudomonas*. Given their unique abilities to consume TPA (*Pseudooceanicola*) and Bd (*Peribacillus*), this DOC removal was plausibly due to consumption of chemicals *Pseudomonas* was unable to metabolize. Taken together, these results imply that complete removal of polymer-derived DOC mobilized by *Pseudomonas* required bacteria capable of metabolizing components that *Pseudomonas* could not. To confirm this, chemical products were tracked using LC−MS.

Tracking Polymer Degradation Products

Representative products were tracked using LC−MS to confirm bacterial transformations, specifically SA, BdTPA, an oxidized form of BdTPA (BdTPAOx), and TPA (Figures 5 and S7 see Supporting Information for conditions where bulk DOC did not accumulate, i.e., those lacking *Pseudomonas*). Note that Bd was not retained on the C18 column and so was outside of the analytical window. In the *Pseudomonas*-only condition, SA accumulated up to 8 h before declining, consistent with bulk DOC measurements and the known ability of *Pseudomonas* to metabolize SA. BdTPA increased to 8 h and then declined as well; however, the concurrent accumulation of BdTPAOx suggested this BdTPA decrease reflected chemical transformation rather than metabolism. Conversion of BdTPA to BdTPAOx is plausible, as oxidation of terminal alcohols to carboxylic acids is a common method by which bacteria metabolize compounds like 1,4-butanediol.[21,62,63] TPA accumulated between 32 and 82 h, indicating TPA-containing products were not metabolized by *Pseudomonas,* as expected. SA and TPA were further quantified using available authentic standards to determine the total contribution each product had to the DOC pool (Figure S8). At 8 h, SA reached its maximum concentration, representing 60.9 ± 4.9% of the DOC and plausibly serving as *Pseudomonas*'s dominant carbon source. In contrast, TPA concentrations peaked much later at 82 h and only made up 3.9 ± 0.2% of the DOC. This low fraction of DOC can be explained by the accumulation of the other TPA-containing products, BdTPA and BdTPAOx. Although these compounds could not be quantified due to the lack of authentic standards, their presence suggests a diverse pool of TPA-containing oligomers formed in the *Pseudomonas*-only incubation. Therefore, the elevated DOC observed in bulk measurements can be explained by the accumulation of these oligomers which *Pseudomonas* was unable to efficiently metabolize or breakdown.

In cocultures with *Pseudooceanicola*, TPA and TPA-containing oligomers were effectively removed over time. Surprisingly, TPA did initially accumulate between 2−12 h in incubations with *Pseudooceanicola*, which likely reflected hydrolysis of oligomeric products into TPA. Such hydrolysis may have temporarily outpaced consumption during early growth, leading to this accumulation. At later time points (32–82 h), TPA was no longer detected, demonstrating consumption of this chemical by *Pseudooceanicola*. Similarly, BdTPA and BdTPAOx did not accumulate, implying these products were broken down and metabolized rather than merely transformed. Together, these results indicate that *Pseudooceanicola* hydrolyzed TPA-containing oligomers and metabolized the released TPA, completing the degradation pathway that *Pseudomonas* alone could not achieve. Interestingly, *Peribacillus* was able to remove TPA-containing compounds as well. Growth assays previously showed this bacterium did not grow on TPA alone, suggesting its ability to process these compounds was either slow or dependent on interactions with other bacteria.

Taken together, the data supports a general scheme for distributed degradation (Figure 6). *Pseudomonas* dominantly degraded the insoluble, high molecular weight polymer presumably via excretion of a PETase-like-enzyme.[12] This mobilization step resulted in dissolved products that were then transformed with specificity by each of the bacteria within the small community (*Pseudomonas*, *Peribacillus*, and *Pseudooceanicola*). Of note, only a fraction of the oligomeric breakdown products were characterized, and further investigations with advanced cheminformatic tools are needed to assign specific bacteria to the transformation of all degradation products. Even with this, tracking the selected products showed *Pseudomonas* performed partial chemical transformations, while *Pseudooceanicola* and *Peribacillus* further degraded TPA-containing products. Based on growth assays (Figure 2) we assigned mineralization of SA, TPA, and Bd to *Pseudomonas* and *Pseudooceanicola*, *Pseudooceanicola*, and *Peribacillus*, respectively. Overall, this work provides direct evidence of the underlying chemical transformations resulting in enhanced mineralization by this bacterial community and highlights the importance of distributed degradation in PBSeT biotransformation.

Implications

This work illustrated the degradation behavior of an environmentally isolated 30-strain bacterial community acting on the aromatic–aliphatic copolyester PBSeT. Most notably, we found that *Pseudomonas* depolymerized the bulk polymer but showed limited ability to degrade soluble, TPA-containing oligomers, whereas *Pseudooceanicola* lacked the ability to depolymerize the polymer but could efficiently break down oligomers into TPA. This division of labor resembles the PETase–MHETase system of *I. sakaiensis*,[32,33,64] showing that this type of distributed degradation can be split among multiple bacteria to mineralize complex copolyesters like PBSeT. Such specialization may help explain why diverse communities populate plastic particles in the ocean, as polymer chemical complexity may select for multiple specialized bacteria participating in distinct steps of polymer mineralization.

In these environmental contexts, reliance on this distributed degradation may challenge natural bacterial communities' ability to degrade these solid materials, as the chance of assembling appropriate metabolic capabilities by planktonic bacteria on solid substrates is fairly low. Importantly, all laboratory experiments in this report occurred in well mixed, sealed cultures, a significantly different environment than what bacteria may experience in nature. For example, in biofilms, such as the ones from which these bacteria were cultured, soluble degradation products would be subject to diffusive and advective dispersal as the particle sinks or is transported through the water column. Therefore, a bacterial community relying on complementary metabolisms would have to form a relatively thick biofilm to prevent diffusive losses and ensure cross feeding of dissolved degradation products.[65] Given these limitations, an important question emerges: how do bacteria access and interact with solid and oligomeric polymer substrates in natural settings? It is well established from environmental field research that bacteria do form biofilms on plastic particles and that unique communities develop in such environments relative to nonplastic particles.[38,43,60,61] These local environments within biofilms may overcome some of the challenges associated with bacterial access to solid substrates. Additionally, species succession within particle-attached communities has been shown to influence degradation,[66] suggesting that biofilm formation may be important for bacteria to liberate high molecular weight substrates for subsequent utilization. Future studies should test whether the complementary metabolisms identified here are general features of biofilm-associated degradation of aromatic aliphatic copolyester plastics.

Overall, this study illustrated that mineralization of aromatic aliphatic copolyesters required the presence of multiple bacteria with specialized metabolisms. A minimum number of bacteria were identified to be sufficient for mineralization, filling the roles of depolymerization and consumption of each polymer component. This highlights that the biotransformation of chemicals in the environment may be controlled by a small subset of specialized organisms that live within the context of more complex environmental communities. Perhaps most critically, this deepened understanding of the microbial ecology in plastic products' end-of-life environment can guide the design of new materials and help constrain the environmental lifetime of existing materials engineered to degrade in the natural world.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.5c14910.

Supporting Information contains further description of isolation procedures, media recipe, serial passaging experiment, raw isotopic data, chromatograms, polyethylene controls, AlphaFold images for protein structure, DOC and LCMS controls, and quantification of SA and TPA (PDF)

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Author Contributions

MJF designed, performed, analyzed lab experiments, and composed the article; CB designed and executed the marine microbial enrichment; DJM helped conduct experiments; AVS and MGH supported isotopic and CO_2 analysis; SG oversaw the microbial consortia enrichment; PAW performed isolation experiments; AS helped design isolation experiments; DLM designed microbial growth experiments and analyzed results; OXC designed and supported microbial isolation experiments; DLP designed experiments, analyzed results, composed the article, and supported the project. All authors edited the paper.

Notes

The authors declare the following competing financial interest(s): S.G. and C.B. are employees of BASF SE. BASF SE produces a polymer used in this study, PBSeT.

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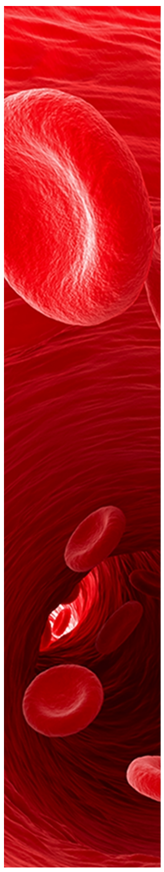

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